Standard for Adhesives for Commercial Use (GS-36)

March 11, 2025

Edition 2.2

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Foreword

General. The final issued standard was developed in an open and transparent process with stakeholder input that included producers, users, and general interests. Edition 2.2 was issued on March 11, 2025. It replaces Edition 2.1 from May 10, 2013. Corrections and/or clarifications were last made to this standard on April 3, 2026. Information on changes made to this standard can be found on Green Seal's website.[1]

The requirements in the standard are based on an assessment of the environmental, health, or social impacts associated with the products, services, or organizations covered in the scope of the standard. These requirements are subject to revision and generally cover aspects above and beyond regulatory compliance. This standard neither modifies nor supersedes laws and regulations. Any conformity assessment to this standard requires compliance with all applicable laws and regulations for the manufacturing and marketing of the products.

Provisions for safety have not been included in this standard since they are supervised by regulatory agencies. Adequate safeguards for personnel and property should be employed for all stages of production and for all tests that involve safety considerations.

Products, services, or organizations that are substantially similar to those covered by this standard in terms of function and life cycle considerations may be evaluated against the intent of the requirements of this standard, accounting for relevant differences between the intended scope of the standard and the actual product, service, or organization to be evaluated.

This standard may not anticipate a feature of the product that may significantly, and undesirably, increase its impact on the environment, health, or society. In such a situation, Green Seal will ordinarily amend a standard to account for the unanticipated environmental, health, or societal impacts.

Normative references (e.g., other standards) in this standard intend to refer to the most recent edition of the normative reference. Test methods may be required for product evaluation. Unless explicitly stated that a specified method is the only acceptable one, the intent of the standard is that an equivalent test method may be accepted, at Green Seal's sole discretion.

Certification to this standard shall be awarded only by Green Seal or, with Green Seal's explicit written permission, by a third-party certification program conducting on-site audits.

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[1] Library of Standards Documents, https://greenseal.org/green-seal-standards/library/

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List of Acronyms and Abbreviations

CARB. California Air Resources Board
CFC. Chlorofluorocarbon
CFR. Code of Federal Regulations
ECVAM. European Centre for the Validation of Alternative Methods
EPA. United States Environmental Protection Agency
IARC. International Agency for Research on Cancer
ICCVAM. Interagency Coordinating Committee on the Validation of Alternative Methods
OECD. Organization for Economic Cooperation and Development
PBT. Persistent, Bioaccumulative, and Toxic Compounds
PFAS. Per- and Polyfluoroalkyl Substances
ppm. Parts per million
SCAQMD. South Coast Air Quality Management District
TG. Testing Guidance
VOC. Volatile Organic Compound

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Green Seal Standard for Adhesives for Commercial Use, GS-36

1.0 Scope

This environmental standard establishes environmental requirements for *adhesives*, *adhesives* applied onto substrates, and *aerosol adhesives*. *Adhesives* covered by the standard shall be intended and labeled for use as a *commercial adhesive*. The standard specifically does not cover *adhesive* tapes or tape strips, textile *adhesives*, optical *adhesives*, *adhesives* used with electronics, packaging *adhesives*, or aerospace *adhesives*. See Appendix 1 for an example list of products included in this standard.

Words and phrases described in the standard that appear in italics have a corresponding definition located in the definition section of the standard, Annex A.

2.0 Safer Chemicals

2.1 Safer Ingredients

2.1.1 Carcinogens. The product shall not be formulated with any *carcinogens*. Any carcinogen that is known to be present as a contaminant shall not exceed 0.1% by weight of the product.

2.1.2 Reproductive Toxins. The product shall not be formulated with any *reproductive toxins*. Any *reproductive toxin* that is known to be present as a contaminant shall not exceed 0.1% by weight of the product.

2.1.3 Per- and Polyfluoroalkyl Substances (PFAS). The product shall not contain any *ingredients* or *components* that are *Per- or Polyfluoroalkyl Substances (PFAS).*

2.1.4 Persistent, Bioaccumulative, and Toxic Compounds. The product shall not be formulated with any *persistent, bioaccumulative, and toxic compounds (PBTs).* Any PBT that is known to be present as a contaminant shall not exceed 0.1% by weight of the product.

2.1.5 Ozone-Depleting Substances. The product shall not be formulated with any *ozone-depleting substances*. Any *ozone-depleting substance* that is known to be present as a contaminant shall not exceed 0.1% by weight of the product.

2.2 Safer Products

2.2.1 Volatile Organic Compounds (VOC) Content. VOCs include all organic compounds that have a vapor pressure of greater than 0.1 mm mercury at 1 atm pressure and 20º C. “VOC content” means the total weight of VOCs in a product expressed as a percentage of the product weight.

The VOC content of the product as applied according to the manufacturer’s recommendations shall not exceed the current regulatory limits of the California Air Resources Board (CARB) for its product category.
The VOC content shall be determined in one of the following ways:

- By summing the percent by weight contribution from all volatile organic components present in the product at 0.01% or more.

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- According to the California Air Resources Board Method 310 (or equivalent), modified to include all volatile organic components present in the product at 0.01% or more[2].

Additionally, for specialty applications not defined by CARB, the current regulatory limits and determination method of the South Coast Air Quality Management District (SCAQMD) Rule 1168 shall be used.

Current CARB regulatory limits for VOCs[3].

Product Category	Effective Date	Limit (%)
Aerosol Adhesives (all types)	1/1/95	75
Mist Spray Adhesive (general purpose)	1/1/2017	30
Web Spray Adhesive (general purpose)	1/1/2017	40
Mounting, Automotive Engine Compartment, and Flexible Vinyl Spray Adhesive	1/1/2002	70
Polystyrene Foam and Automobile Headliner Spray Adhesive	1/1/2002	65
Polyolefin and Laminate Repair/Edgebanding Spray Adhesive	1/1/2002	60
Construction, Panel, and Floor Covering	12/31/2008	7
Contact Adhesive – General Purpose	12/31/2006	55
Contact Adhesive – Special Purpose	12/31/2006	80
General Purpose Adhesives	1/1/95	10

Current SCAQMD regulatory limits for VOCs[4].

Product Category	Effective Date	Limit (%)
PVC Welding	1/1/2005	51
CPVC Welding	1/1/2005	49
ABS Welding	7/1/2005	32
Plastic Cement Welding	1/1/2005	25

[2] Evaluation of total VOCs in this standard includes all volatile organic compounds present in the product at 0.01% or more. Evaluation of total VOCs under Method 310 exempts fragrances and all organic compounds present below 0.1%.
[3] These limits are a reference to the current CARB regulatory limits and will be updated to reflect any amendments made by CARB in the future.
[4] These limits are a reference to the current SCAQMD regulatory limits and will be updated to reflect any amendments made by SCAQMD in the future.

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Product Category	Effective Date	Limit (%)
Adhesive Primer for Plastic	7/1/2005	55
Computer Diskette Manufacturing	1/1/2005	35
Special Purpose Contact Adhesive	1/1/2005	25
Tire Retread	1/1/2005	10
Adhesive Primer for Traffic Marking Tape	1/1/2005	15
Structural Wood Member Adhesive	1/1/2005	14
Sheet Applied Rubber Lining Operations	1/1/2005	85
Top and Trim Adhesive	1/1/2007	25

2.2.2 Toxic Compounds. The solvent portion of the *adhesive* shall not be toxic to humans when inhaled. A product is considered toxic if the following lethal dose (LD) criterion applies:

Inhalation $LC_{50} \leq 2{,}000$ parts per million (ppm) of vapor or gas or 20 mg/L of mist, dust, or fumes

The toxicity testing procedures shall follow the protocols put forth in the Organization for Economic Cooperation and Development (OECD) Guidelines for the Testing of Chemicals, which includes: Acute Inhalation Toxicity Test (Testing Guidance (TG) 403). To demonstrate compliance with this requirement, a solvent need not be tested if existing toxicological information demonstrates that it complies. Data from the Registry of Toxic Effects of Chemical Substances and from the Hazardous Substances Data Bank will be accepted as well as peer-reviewed primary data.

2.2.3 Animal Testing. To avoid new animal testing, previous test results will be accepted as evidence of meeting a criterion. When existing data are not available, the preferred methods for new testing include methods that replace, reduce, or refine animal use, particularly those recommended by the Interagency Coordinating Committee on the Validation of Alternative Methods (ICCVAM) or the European Centre for the Validation of Alternative Methods (ECVAM), unless indicated otherwise. In addition, other non-animal (in-vitro) test results, modeling data, data from structural analogs, and other lines of evidence may be accepted, provided that the methods are peer-reviewed and applicable. Specific in vitro or modeling methods may be noted in the standard, but additional options may be accepted by the certification program.

Further, a mixture need not be tested if existing information demonstrates that each of the applicable *components* complies with the criterion.

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3.0 Sustainable Packaging

3.1 Packaging Materials.

3.1.1 Packaging. Product packaging shall be resealable after the first use except for single-use packaging. Shipping packaging shall be reusable, recyclable or reconditionable. Corrugated shipping packaging shall contain 30% minimum postconsumer recycled content.

4.0 Verified Performance and Claims

4.1 Product Performance. Product shall be approved by the manufacturer for use as an *adhesive* type outlined in Section 1.0.

5.0 Trademark Use Requirements

5.1 Trademark Use. Any use of the Green Seal® Certification Mark or the Green Seal name, e.g., on the product, product label, packaging, secondary documents, or promotional materials, must be in accordance with Green Seal's Trademark Use Guidelines.[5]

5.2 Misleading Claims. Green Seal trademarks shall not be used in conjunction with any modifying terms, phrases, or graphic images that might mislead consumers as to the extent or nature of the certification.

[5] www.greenseal.org/trademark-use-guidelines

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Annex A – Definitions (Normative)

Note that the defined terms are italicized throughout the standard.

Adhesive. A substance capable of holding materials together by surface attachment.

Carcinogens. A chemical listed as known, probable, reasonably anticipated, or possible human carcinogen by and of the following agencies or programs, with the exception of crystalline silica:

- The International Agency for Research on Cancer (IARC) – Groups 1, 2A, and 2B

Commercial Adhesive. An *adhesive* product that is designed for use in the maintenance or operation of an establishment that manufactures, transports, or sells goods or commodities, or provides services for profit; or is engaged in the nonprofit promotion of a particular public, educational, or charitable cause. Establishments include, but are not limited to, government agencies, military organizations, factories, schools, hospitals, sanitariums, prisons, libraries, office complexes, restaurants, hotels, stores, automobile service and parts centers, health clubs, theaters, or transportation companies. *Commercial Adhesives* do not include products that are incorporated into or used exclusively in the manufacture or construction of the goods or commodities that are produced by the establishment.

Component. A constituent that is deliberately added at any level for its continued presence in the final product to provide a specific characteristic, appearance, or quality.[6]

Ingredient. Any constituent of a product that is intentionally added or known to be a contaminant that comprises at least 0.01% by weight of the product.

Ozone-Depleting Substances. A compound with an ozone depletion potential greater than 0.01 (Chlorofluorocarbon (CFC) 11=1), according to the United States Environmental Protection Agency (EPA) list of Class I and Class II Ozone Depleting Substances.

Per- and Polyfluoroalkyl Substances (PFAS). A class of fluorinated organic chemicals containing at least one fully fluorinated carbon atom.

Persistent, Bioaccumulative, and Toxic Compounds. Compounds identified by the EPA Waste Minimization Branch as being persistent, bioaccumulative, and toxic (http://www.epa.gov/minimize/chemlist.htm).

Reproductive Toxins. A chemical listed as a reproductive toxin (including developmental, female, and male toxins) by the State of California under the Safe Drinking Water and Toxic Enforcement Act of 1986 (California Code of Regulations, Title 22, Division 2, Subdivision 1, Chapter 3, Sections 1200 et. Seq., also known as Proposition 65).

[6] Naturally occurring elements and chlorinated organics that may be present as a result of chlorination of the water supply, are not considered intentional components if the concentrations are below the applicable maximum contaminant levels in the National Primary Drinking Water Standards found in 40 CFR Part 141.

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Appendix 1 – Scope (Informative)

Examples of products included in or excluded from the scope of GS-36:

Products Included in GS-36

- Acrylonitrile-butadiene-styrene welding *adhesive*
- Carpet pad installation *adhesive*
- Ceramic tile installation *adhesive*
- Contact bond *adhesive*
- Contact bond-specialty substrates *adhesive*
- Cove base installation *adhesive*
- Chlorinated polyvinyl chloride welding *adhesive*
- Indoor floor covering installation *adhesive*
- Multipurpose construction *adhesive*
- Nonmembrane roof installation/repair *adhesive*
- Other plastic cement welding *adhesive*
- Outdoor floor covering installation *adhesive*
- Polyvinyl chloride welding *adhesive*
- Rubber floor installation *adhesive*
- Single-ply roof membrane installation/repair *adhesive*
- Structural glazing *adhesive*
- Perimeter bonded sheet vinyl flooring installation *adhesive*
- Waterproof resorcinol glue
- Wood flooring *adhesive*

Products Excluded from GS-36

- Aerospace *adhesives*
- Consumer/office *adhesives*
- Electronic *adhesives*
- Label *adhesives*
- Office *adhesives*
- Optical *adhesives*
- Packaging *adhesives*
- Tape *adhesives*
- Textile *adhesives*

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